

## Secant Equation (0.5 pt)

The secant equation is:

$$B_{k+1}s_k = y_k,$$

where  $s_k = x_{k+1} - x_k$ ,  $y_k = \nabla f_{k+1} - \nabla f_k$ , and  $B_{k+1}$  is an approximation of the Hessian  $\nabla^2 f_{k+1}$ . Prove that the secant equation is consistent with the minimization of the quadratic model problem  $m(v)$ .

**Solution.** Consider the standard quadratic model around  $x_k$

$$m(v) = f(x_k) + \nabla f(x_k)^\top v + \frac{1}{2}v^\top B_{k+1}v, \quad v \in \mathbb{R}^n.$$

Its gradient is

$$\nabla m(v) = \nabla f(x_k) + B_{k+1}v.$$

To make the model consistent with the true function at the next iterate  $x_{k+1} = x_k + s_k$ , we require that the model gradient interpolates the true gradient at both  $v = 0$  and  $v = s_k$ :

$$\nabla m(0) = \nabla f(x_k), \quad \nabla m(s_k) = \nabla f(x_{k+1}).$$

The second condition gives

$$\nabla f(x_k) + B_{k+1}s_k = \nabla f(x_{k+1}) \iff B_{k+1}s_k = \nabla f(x_{k+1}) - \nabla f(x_k) = y_k,$$

which is precisely the secant equation.

## Curvature Condition (0.5 pt)

Show that the curvature condition  $s_k^\top y_k > 0$  is required for the Quasi-Newton method. Prove that the curvature condition is satisfied when the line search algorithm for the Wolfe conditions is used.

**Solution.** (i) **Derivation for a general quadratic model.**

To make the quadratic model seen before consistent with the observed change in gradient along the realized step  $s_k = x_{k+1} - x_k$ , the secant equation  $B_{k+1}s_k = y_k$  must hold. We assume that a positive definite  $B_{k+1}$  exists, then

$$s_k^\top y_k = s_k^\top B_{k+1}s_k > 0,$$

since  $B_{k+1} \succ 0$  implies  $z^\top B_{k+1}z > 0$  for all nonzero  $z$ . Therefore, if a convex quadratic model consistent with the step exists, the curvature condition  $s_k^\top y_k > 0$  is necessary.

(ii) **Wolfe conditions.** Let  $\Delta x$  be a descent direction at  $x$  and define

$$\phi(t) = f(x + t\Delta x), \quad \phi'(t) = \nabla f(x + t\Delta x)^\top \Delta x.$$

Suppose  $t > 0$  is chosen by a line search that enforces the Wolfe conditions with  $\alpha \in (0, 1)$  and  $\beta \in (\alpha, 1)$ :

$$\begin{aligned} \text{(sufficient decrease)} \quad & f(x + t\Delta x) \leq f(x) + t\alpha \nabla f(x)^\top \Delta x & \iff & \phi(t) \leq \phi(0) + t\alpha \phi'(0), \\ \text{(curvature)} \quad & \nabla f(x + t\Delta x)^\top \Delta x \geq \beta \nabla f(x)^\top \Delta x & \iff & \phi'(t) \geq \beta \phi'(0). \end{aligned}$$

Set  $s_k = t\Delta x$  and  $y_k = \nabla f(x + t\Delta x) - \nabla f(x)$ . Then

$$s_k^\top y_k = t(\Delta x^\top \nabla f(x + t\Delta x) - \Delta x^\top \nabla f(x)) = t(\phi'(t) - \phi'(0)).$$

By the curvature Wolfe condition,  $\phi'(t) \geq \beta \phi'(0)$ . Since  $\Delta x$  is a descent direction,  $\phi'(0) = \nabla f(x)^\top \Delta x < 0$ . Therefore,

$$s_k^\top y_k \geq t(\beta \phi'(0) - \phi'(0)) = t(\beta - 1) \phi'(0) > 0,$$

because  $\beta - 1 < 0$ ,  $\phi'(0) < 0$ , and  $t > 0$ . Hence any step length chosen under these Wolfe conditions satisfies the curvature condition  $s_k^\top y_k > 0$ .